**6** $A$ is the point with coordinates (5, 9)
$B$ is the point with coordinates ($d$, 15)

The gradient of the line $AB$ is 3

Work out the value of $d$.

$$\text{Gradient} = \frac{15 - 9}{d - 5} = 3$$

**1 mark**
Use change in y ÷ change in x

$$6 = 3(d - 5)$$
$$\div 3 \qquad \div 3$$
$$2 = d - 5$$
$$+5 \qquad +5$$
$$7 = d$$

**1 mark**
Rearrange to isolate d

7

**1 mark**
d = 7

**(Total for Question 6 is 3 marks)**

DO NOT WRITE IN THIS AREA
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